

Ayman Mahfuz

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EDUCATION

The University of Texas at Austin

Aug 2023 - May 2027

Bachelor of Science, Computer Science, Concentration: AI and ML

Austin, TX

• GPA: 3.5/4.0

• Coursework: High Performance Computing, Natural Language Processing, Generative Visual Computing, Energy-Efficient Computing

EXPERIENCE

Arm

May 2025 - Aug 2026

Platform Validation Intern / Co-op – Software Engineering

- Built and deployed a Python Bayesian optimizer over a 63-dimensional stress-ng configuration space to reproduce silicon failures across three CPU-platform validation phases, avoiding an estimated 10,000+ hours of exhaustive testing.
- Instrumented ML-guided stress testing with CPU performance counters for cache misses, memory bandwidth, and stall rate, directing Bayesian search toward high-stress configurations, which accelerated identification of silicon-failure cases and shortened validation cycles
- Earned internal patent-filing approval and a 12-month co-op extension; presented production results to SVP and director leadership.

UT Austin – Texas Robotics & AI Lab

Jan 2025 - Jul 2026

Undergraduate Research Assistant – Robotics RL & LLM Evals

- Built a reproducible multi-turn LLM evaluation harness measuring whether an RL advisor trained on a coin game transfers to persuasion dialogues (Persuade for Good dataset); served three Qwen variants via vLLM on Texas Advanced Computing Center GPUs and found advisor raised simulated donation rate by 30%.
- Accelerated policy-training iterations 2–3x by profiling GPU utilization and removing training-pipeline bottlenecks; scaled experiments to 5M+ episodes.
- Deployed 15 hierarchical reinforcement-learning policies to NAO robots and built C++/Python training environments within a 400K+ line codebase, contributing to the team's third-place finish at RoboCup 2025.

UT Austin – Center for Media Engagement

Aug 2023 - May 2025

Undergraduate Research Assistant – Software Engineering

- Built concurrent Selenium and REST ingestion workers with retry handling, enabling reliable processing of a 250M+ row BigQuery dataset containing 50M+ articles and 70M+ comments, which reduced ingestion failures and accelerated pipeline throughput.
- Prevented duplicate experiment events and preserved round-robin assignments for 1,000+ participants through deterministic assignment logic, event deduplication, and structured logging.

UT Austin – Oden Institute

Aug 2023 - Jan 2025

Undergraduate Research Assistant – ML Systems

- Improved MRI segmentation Dice score by 12 percentage points (from 0.70 to 0.82) by benchmarking CNN and transformer architectures U-Net, MedSAM, and PanSegNet with five-fold cross-validation on 1,000+ scans using NVIDIA H100 GPUs

PROJECTS

HelmPM - Multi-Tenant Project Management Platform

- Built and launched a multi-tenant project management platform used by 5 organizations to manage 5,000+ tickets, with a React frontend, FastAPI backend, PostgreSQL row-level tenant isolation, and bidirectional Linear/Jira sync.
- Eliminated re-embedding of unchanged documents with Merkle-tree indexing, reducing API costs by 40%; implemented semantic, keyword, and exact-ID search across documents and code.

Activation Checkpoint Integrity in PyTorch - Independent Research

Python, PyTorch, CUDA

- Built a PyTorch integrity checker that detects silent numerical divergence in activation checkpointing by pairing original and recomputed tensors, localizing faults to specific layers, and blocking optimizer steps on mismatch, with 1.8% runtime overhead, conducted under Professor Mootaz Elnozahy.

Energy Profiling of LLM & Diffusion Inference

Python, PyTorch, vLLM, SLURM

- Ran 577 GH200 experiments across nine models and 55 configurations, measuring 24–32% lower LLM inference energy with FP8/INT8 and no measurable diffusion savings; built a vLLM/SLURM harness instrumented with NVML/Zeus and analyzed results using Welch's t-tests, Bonferroni correction, and roofline analysis.

Transformer Pretraining & Post-Training Pipeline (253M Parameters)

PyTorch, HPC

- Pretrained a 253M-parameter decoder-only transformer (RoPE, RMSNorm, SwiGLU) on 600M tokens, using a resumable bf16 mixed-precision training pipeline on HPC clusters with checkpoint recovery; extended it with SFT and DPO post-training.

SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, C

ML: PyTorch, Transformers, vLLM, LoRA/PEFT, PPO/MAPPO, RLHF/DPO, MCP, Harness Engineering

Systems/Cloud: CUDA, SLURM, NVML/Zeus, AWS, Docker, Linux, Distributed Systems

Backend/Web/Data: React, FastAPI, Flask, PostgreSQL, BigQuery, REST APIs, Pandas, NumPy